



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No.: 04

Title: Use function types, scope, and closures; apply try-catch; build a palindrome checker.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
let welcome = function() {  
    console.log("Welcome to the Palindrome Checker!");  
};  
  
welcome();  
  
// Global variable  
let language = "JavaScript";  
  
// Main function to check if a string or number is a palindrome  
function isPalindrome(str) {  
    // 1. Accept numbers by converting them to strings  
    if (typeof str === "number") {  
        str = String(str);  
    }  
  
    // 2. Validate input type (reject null, undefined, objects, arrays, etc.)
```



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```
if (typeof str !== "string") {  
    throw new TypeError(`Expected a string or number, but received ${typeof str}`);  
}  
  
// 3. Check for empty string  
if (str.trim().length === 0) {  
    throw new Error("Input string cannot be empty.");  
}  
  
// Clean the string: remove non-alphanumeric characters and convert to lowercase  
let cleaned = str.toLowerCase().replace(/[^a-z0-9]/g, "");  
  
// Handle edge case where input had only punctuation/spaces (e.g., "!!!")  
if (cleaned.length === 0) {  
    throw new Error("Input must contain at least one alphanumeric character.");  
}  
  
// Reverse and check  
let reversed = cleaned.split("").reverse().join("");  
return cleaned === reversed;  
}
```



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// Helper function with try...catch error handling

```
function testPalindrome(phrase) {
```

```
  try {
```

```
    let result = isPalindrome(phrase);
```

```
    console.log(` SUCCESS: "${phrase}" -> ${result ? "Is a palindrome!" : "Not a  
palindrome."}`);
```

```
  } catch (error) {
```

```
    // Handle caught errors safely without stopping execution
```

```
    console.error(` ERROR [${error.name}]: Failed to check input (${phrase}) ->  
${error.message}`);
```

```
  }
```

```
}
```

// --- Testing Examples ---

```
console.log(`\nRunning checks using ${language}:\n` + "=".repeat(45));
```

// Valid Test Cases (Strings & Numbers)

```
testPalindrome("Hello World");
```

```
testPalindrome("Aishwarya Brahmkar");
```

```
testPalindrome("24070521212");
```

```
testPalindrome(12321);           // NOW WORKS: Passed as a Number directly!
```

```
testPalindrome(21212);           // NOW WORKS: Evaluates as "Not a palindrome"
```



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// Error Handling Test Cases (Triggers Catch Block)

testPalindrome(""); // Error: Empty string

testPalindrome(null); // TypeError: null passed

testPalindrome(" "); // Error: Whitespace only

testPalindrome("!!!?"); // Error: No alphanumeric characters

Output:

```
[Running] node "e:\JS lab\Experiment_4\palindrome_checker.js"
Welcome to the Palindrome Checker!
ERROR [Error]: Failed to check input () -> Input string cannot be empty.
ERROR [TypeError]: Failed to check input (null) -> Expected a string or number, but received object
ERROR [Error]: Failed to check input ( ) -> Input string cannot be empty.

Running checks using JavaScript:
=====
SUCCESS: "Hello World" -> Not a palindrome.
SUCCESS: "Aishwarya Brahmkar" -> Not a palindrome.
SUCCESS: "24070521212" -> Not a palindrome.
SUCCESS: "12321" -> Is a palindrome!
SUCCESS: "21212" -> Is a palindrome!
ERROR [Error]: Failed to check input (!!!?) -> Input must contain at least one alphanumeric character.

[Done] exited with code=0 in 0.158 seconds
```



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Case Study No.: 04

Title: Use function types, scope, and closures; apply try-catch; build a palindrome checker.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
const readline = require('readline');
```

```
// 1. FUNCTION DECLARATION (Global Scope)
```

```
// Utility function to reverse the entered PIN
```

```
function reversePin(pin) {  
    const pinStr = String(pin);  
    return pinStr.split("").reverse().join("");  
}
```

```
// Helper function to ask a question in the terminal using built-in Node readline
```

```
const askQuestion = (query) => {  
    const rl = readline.createInterface({  
        input: process.stdin,  
        output: process.stdout  
    });  
    return new Promise(resolve => rl.question(query, answer => {
```



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```
    rl.close();

    resolve(answer);

  });
};

// 2. CLOSURE & SCOPE DEMONSTRATION

// ATM Session Generator returning an Arrow Function

const createATMSession = (accountHolder) => {

  // Outer Lexical Scope variable

  const user = accountHolder;

  let attempts = 0;

  // 3. ARROW FUNCTION (Closure over `user` and `attempts`)

  const verifyPinInteractive = async () => {

    attempts++;

    // Dynamic prompt to ask the user for PIN input

    const enteredPin = await askQuestion(`[ATM System] ${user}, please enter any PIN: `);
```



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```
if (!enteredPin || enteredPin.trim() === "") {  
    return `Transaction cancelled by ${user}.`;   
}  
  
const strEnteredPin = String(enteredPin.trim());  
const reversed = reversePin(strEnteredPin); // Calls global Function Declaration  
  
console.log(`\n--- Transaction Attempt #${attempts} for ${user} ---`);  
console.log(`Entered PIN : ${strEnteredPin}`);  
console.log(`Reversed PIN: ${reversed}`);  
  
// Check if the entered PIN is a Palindrome  
const isPalindrome = (strEnteredPin === reversed);  
  
// Access is ALWAYS granted, but security message depends on Palindrome check  
if (isPalindrome) {  
    return `ACCESS GRANTED!\n SPECIAL SECURITY ALERT: PIN "${strEnteredPin}"  
is a Palindrome! Please consider changing your password for better security.`;  
} else {  
    return `ACCESS GRANTED! Welcome, ${user}.`;   
}  
};
```



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```
return verifyPinInteractive;

};

// 4. ATM SYSTEM EXECUTION

(async () => {

    const johnSession = createATMSession("John Doe");

    // Prompts user in the integrated terminal

    const result = await johnSession();

    console.log("\n" + result);

})();
```

Output:

```
PS E:\JS lab> node "e:\JS lab\Aishwarya_Brahmankar_24070521212\Case Study 4\ATM_pin_verification.js"
[ATM System] John Doe, please enter any PIN: 1221

--- Transaction Attempt #1 for John Doe ---
Entered PIN : 1221
Reversed PIN: 1221

ACCESS GRANTED!
SPECIAL SECURITY ALERT: PIN "1221" is a Palindrome! Please consider changing your password for better security.
```